

⇒ ML: Teach computer to learn from data!

→ Supervised Learning = Learning with labeled data.

Types of Supervised Learning:

1. Regression.

2. Classification.

→ Regression: predict a continuous number.

ex: predict house price

→ input: Area, bedrooms, location

→ output: price (like \$50, \$80, etc)

→ Model learn pattern.

Common regression types:

→ Linear Regression (straight line prediction)

→ Polynomial Regression (curved line prediction)

4 Classification:-

Predict a (category/label)

ex: Email spam filter

common classification types!

→ Binary (2 classes: Yes/no, spam/not spam)

→ Multi-class (multiple classes: cat/dog/bird in image)

Regression predicts continuous numbers
like (rice 50.5L, 80.2L)

→ Classification predicts categories like
spam (not spam)

Classification algorithms predict!

→ Discrete categories (like spam/not spam)
(cat/dog)

- Not continuous numbers.

Discrete = separate categories!

~ like labels: "Red", "Green", "Blue"

~ Not like numbers: 1.5, 2.2, 3.14

Ex:-

- 1. Logistic Regression (binary classification)
- 2. Decision Tree (Multi-class)
- 3. SVM

- 4, Random Forest
- 5, KNN.
- 6, Naive Bayes

⇒ Unsupervised Learning = Learning without labels!

2 Main Types of Unsupervised Learning!

1 Clustering!

Combine similar data points.
ex: Think of it is like making groups of similar habits.

~ Algorithms

i, k-means: Divide data into k groups

~ ii, Hierarchical clustering → Tree like structure

2/ Association / Dimensionality Reduction:

This type is divided into the two subparts!

A. Association: Rule Learning!
Finding Rules, like if this then this will happen
ex: Bread + cheese always often come together

B. Dimensionality Reduction!

Reducing non-important features and convert it into few features.

also instants:

1/ clustering $\rightarrow$ Grouping

2/ Association $\rightarrow$ finding relationships / rules.